PRODUCT OVERVIEW

Monad Validator API

Real-time validator metadata, uptime tracking & on-chain staking data

Node.js + Express

SQLite WAL

Epoch-Based Indexing

Live RPC

REST API

Core Features

Everything needed to monitor and interact with Monad validators

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Validator Metadata

Auto-synced from GitHub hourly. Names, logos, websites, social links for all validators.

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Real-Time Uptime

Live log analysis from journalctl. Tracks finalized blocks and timeout events per round, per validator.



Historical Uptime

Epoch-based snapshots with cumulative views: 30-day, 90-day, and all-time uptime percentages.

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On-Chain Staking

Live queries to the staking precompile. Stake amounts, commission rates, unclaimed rewards.

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Delegation Data

Per-validator delegator lists and per-address delegation summaries with reward tracking.

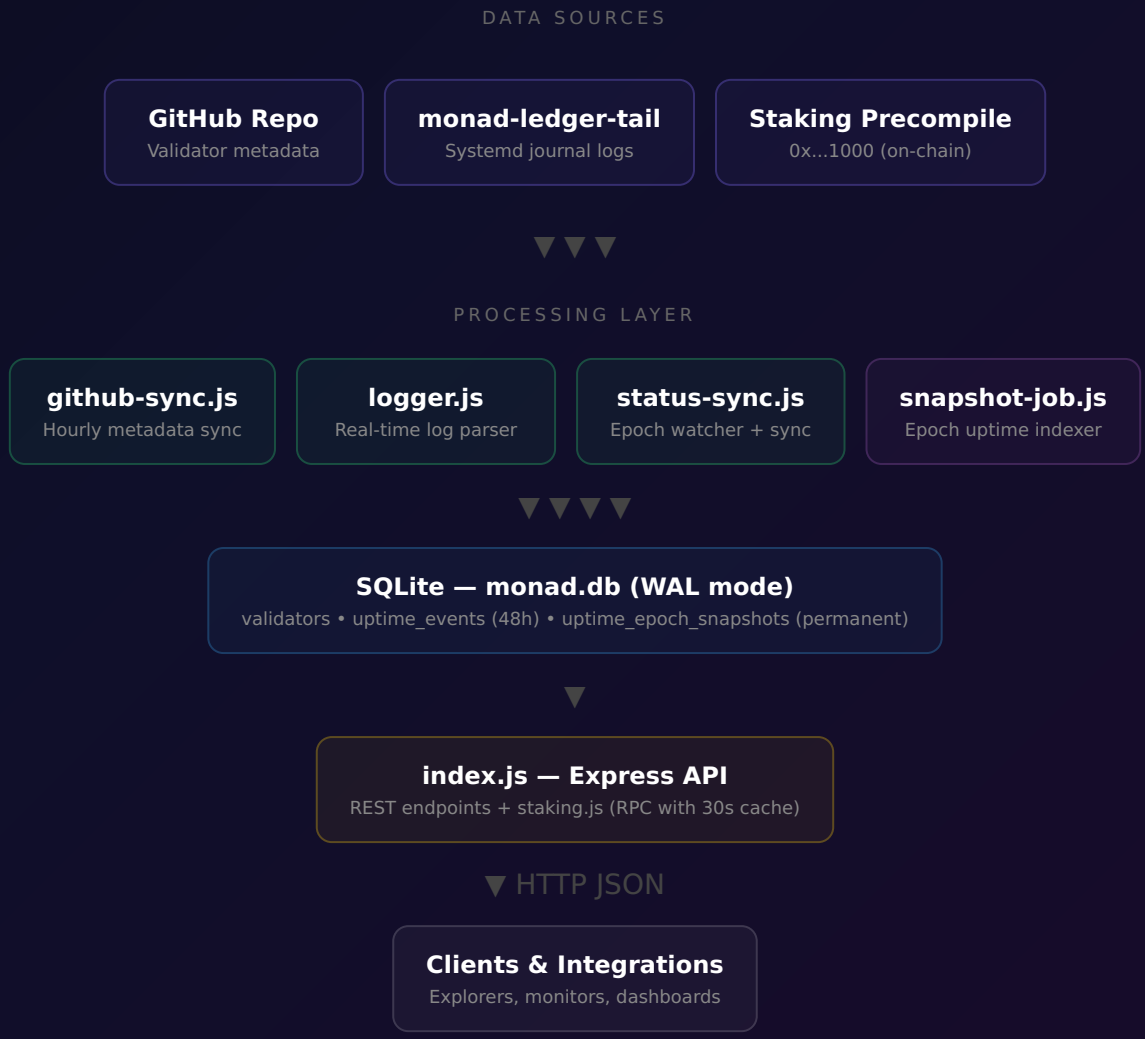
⚙

Epoch Awareness

Polls every 2 minutes. Detects epoch transitions and boundary blocks, triggers full sync automatically.

System Architecture

Two systemd services sharing a single SQLite database (WAL mode)



Epoch-Based Uptime Indexing

Historical uptime tracking with zero data loss guarantees

Real-Time Layer

Validator node logs are parsed in real-time via journalctl. Every finalized block and timeout event is captured per round, per validator.

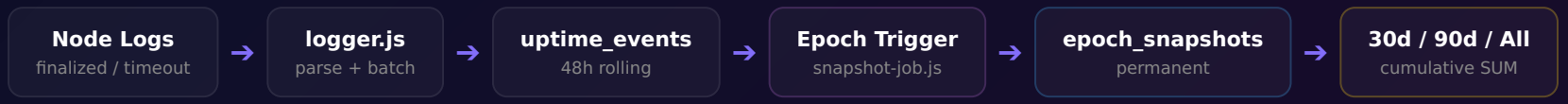
`uptime_events`	48-hour rolling window
Insert rate	~1 second batch flush
Deduplication	UNIQUE(validator_id, round)
Used by	24h live uptime API

Historical Layer

At every epoch transition (~5.5h), all events since the last snapshot are aggregated and permanently stored per validator.

`uptime_epoch_snapshots`	Permanent storage
Trigger	Epoch transition event
Granularity	Per epoch, per validator
Queries	30d / 90d / all-time SUM

DATA FLOW



Safety guarantees: Events are never deleted before being snapshotted. If no snapshot exists, cleanup is blocked. The epoch watcher runs independently, ensuring snapshots are captured even across server restarts.

API Endpoints

RESTful JSON API — Base URL: /monad-api

Validator Endpoints

METHOD	ENDPOINT	DESCRIPTION
GET	/validators	All validators (metadata + stake)
GET	/validators/:id	Single by ID or secp address
GET	/validators/uptime	All uptime (?window, ?sort)
GET	/validators/uptime/:id	Single validator uptime
GET	/validators/uptime/:id/history	Round-by-round history
GET	/validators/stats	Validator count

Staking Endpoints (Live RPC)

METHOD	ENDPOINT	DESCRIPTION
GET	/staking/epoch	Current epoch + delay status
GET	/staking/pool	Bonded tokens, total staked
GET	/staking/validator/:id	Stake, commission, rewards
GET	/staking/validator/:id/delegators	All delegators of validator
GET	/staking/delegator/:addr	Delegations for address

System

GET	/health	Health check

Technical Overview

Lightweight, single-machine deployment with zero external dependencies

Tech Stack

COMPONENT	TECHNOLOGY
Runtime	Node.js 18+ (ESM)
HTTP Framework	Express 4.x
Database	SQLite via better-sqlite3 (WAL)
Blockchain	ethers.js 6.x (RPC + ABI)
Log Source	journalctl (systemd)
Process Manager	systemd (2 services)

Performance Characteristics

METRIC	VALUE
Event insert latency	~1ms (batch, WAL)
API response time	< 10ms (cached)
RPC cache TTL	30 seconds
Memory usage	~40 MB per service
Epoch poll interval	120 seconds
Metadata sync	Hourly (GitHub)

Database Tables

TABLE	PURPOSE	RETENTION
`validators`	Metadata + staking	Permanent
`uptime_events`	Live round events	48 hours
`uptime_epoch_snapshots`	Historical uptime	Permanent

Deployment

SERVICE	PROCESS
`monad-api`	API + sync + snapshots
`monad-uptime-logger`	journalctl parser

Both services share the same SQLite database via WAL mode, enabling concurrent reads and writes without locking.

Ecosystem Integrations

Projects and tools that consume Monad Validator API data

EXPLORER

Monad Scope

monad-scope-piki-nodes.com

Comprehensive Web3 analytics dashboard for the Monad network. Real-time network activity monitoring, block proposer tracking, TPS metrics, and validator performance analytics with an interactive global node map.

- Network Dashboard
- Validator Analytics
- Global Node Map
- Staking Portal

EXPLORER

Unity Nodes Pulse

pulse.unitynodes.com

Professional validator monitoring and management platform by Unity Nodes. Multi-network support with Prometheus + Grafana stack for collecting and visualizing validator metrics across 10+ blockchain networks.

- Multi-Network
- Prometheus
- Grafana
- Professional Ops

OPEN-SOURCE TOOL

Micto Monad Monitor

github.com/MictoNode/micto-monad-monitor

Remote monitoring solution for Monad validators. Cross-validates status using Huginn API, provides multi-channel alerts (Telegram, Discord, Pushover), web dashboard, and 6-hour detailed uptime reports.

- Huginn API Integration
- Telegram Alerts
- Web Dashboard
- Self-Hosted

Open data infrastructure: Monad Validator API serves as a foundational data layer for the Monad ecosystem. Third-party explorers, monitoring tools, and dashboards rely on its endpoints for real-time validator performance data, staking information, and network state.

SUMMARY

Monad Validator API

Production-grade validator infrastructure for the Monad ecosystem

200+	~5.5h	<10ms	3
VALIDATORS TRACKED	EPOCH SNAPSHOTS	API RESPONSE	ECOSYSTEM USERS

Real-Time Uptime · Historical Indexing · On-Chain Staking · Delegation Data · Epoch Awareness · Zero Data Loss

github.com/s0urledd/monad-validator-api